

CSC311 Machine Learning Challenge

GenAI Usage Disclosure

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This disclosure covers only the genAI usage by Yanming Zhao and does not represent the practices of other group members.

Tool Used

Tool: Claude Code (claude-sonnet-4-6), developed by Anthropic.

Usage scope: Code generation for the Random Forest component of this project.

What Was Generated

Claude Code was used to generate two Python files related to the Random Forest model:

- `train_rf.py` — Training pipeline including data loading, hyperparameter grid search over `n_estimators`, `max_depth`, and `max_features` using 5-fold stratified cross-validation, confusion matrix and feature importance plots, and model export via `joblib`.
- `pred_rf.py` — Inference script applying the same preprocessing pipeline as the rest of the project, loading the trained model, and implementing the required `predict_all(filename)` interface.

What Was My Own Work

The decision to use Random Forest as the model for my component was my own, based on the nature of the dataset (mixed numerical and one-hot features, non-linear relationships between emotional responses and painting classes) and my understanding of the tradeoffs between model families discussed in class.

The hyperparameter search space (values of `n_estimators`, `max_depth`, and `max_features`) was determined by me based on common practices and the dataset size. The interpretation of the grid search results and the conclusion that `n_estimators=300`, `max_depth=None` achieved the best CV accuracy of 88.15% was verified and understood by me.

How GenAI Was Used

I provided Claude Code with the context of the existing project (data format, preprocessing pipeline, and the structure of `train_lr.py` and `pred_lr.py`) and instructed it to

generate analogous files for Random Forest. I reviewed, ran, and debugged the generated code (e.g., fixing a Unicode encoding error on Windows), and verified the output results.

No genAI was used to generate any written (English) components of the main report `csc311challenge.pdf`.